Investigation of Guided Recess and Executive Function in Elementary School Children

Emily O. Prager and Stephanie M. Carlson
University of Minnesota-Institute of Child Development

Background

- Playtime in schools is decreasing in favor of increased instructional time (Hofferth & Sandberg, 2001)

- There is a relation between individual differences in play and stronger executive function (e.g., Carlson, White, & Davis-Unger, 2014)
  - Prefrontal cortex function
  - Improved cognitive flexibility
  - Increased creativity

- Guided recess program has been developed to promote these activities
  - Full-time coaches
  - Encourage organized and energetic play at recess
  - Multiple games to choose from
  - Conflict resolution
  - Student leadership
  - Low-income elementary schools
  - Reported increased self-esteem and school attendance and decreased bullying (Bleeker et al., 2012)

Hypothesis

- Possible benefit of guided recess is to promote EF through increased physical activity and adult scaffolding of play

- We hypothesized that children who were exposed to guided recess would show improved EF abilities

Many thanks to the research assistants involved in testing, the members of the Developmental Social Cognitive Neuroscience Lab , and to the children and schools who participated in this research. Funding: NSF Doctoral Fellowship Grant #00039202. **Contact: oste0317@umn.edu**

Methods

Participants

	Kindergarten	2nd Grade	Free/ Reduced Lunch
Guided Recess	17	12	>50%
Typical Recess	12	12	>50%

Dependent Measures

- NIH Toolbox Vocabulary Test
- NIH Toolbox Dimensional Change Card Sort (DCCS)
- Head, Toes, Knees, and Shoulders (HTKS)
- Walking Subtest - Movement Assessment Battery for Children



Figure 1. Sample Images of Vocabulary Test and DCCS

Testing Procedure

- Testing time points: Winter and Spring
- Testing conducted simultaneously at both schools, two week time period (5 days at each school)
- Testing done individually (~15 mins)

Results

- Significant main effects of Time across grades
  - HTKS: $F(1, 48) = 6.49$ $p = .01$
  - Walking: $F(1, 48) = 6.78$, $p = .01$
- No significant main effects of Recess type were found for either grade
- Among the 2nd graders there was a trending significant Time x Recess interaction
  - $F(1, 21) = 4.087$ $p = .056$ (Fig. 2).

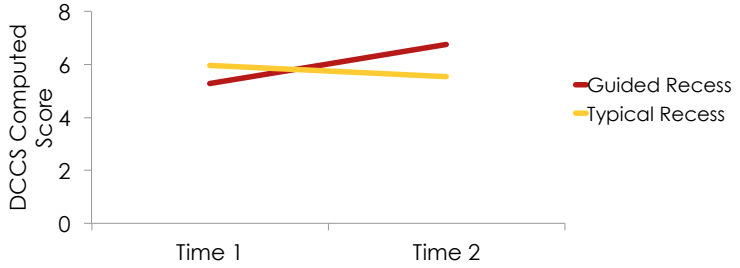


Figure 2. DCCS Computed Scores for 2nd Graders from Time 1 to Time 2

Discussion

- Children across grades in both recess programs improved over the study period
- Second graders showed greater improvement on the DCCS among children receiving guided recess
  - This could indicate that a certain dosage of guided recess must be reached before its impact is seen since the older children had increased exposure
  - This also suggests that the skills promoted in the guided recess program seem to specifically target children's flexible thinking such as task and rule switching
    - This may be especially true for the older children engaging in the guided recess program

Bleeker, M., James, Burdumy, S., Beyler, N., Dodd, A.H., London, R.A., Westrich, L., Stokes-Guinan, K., Castrechini, S. (2012). *Findings from a randomized experiment of Playworks: Selected results from cohort 1.* (Research Report). Retrieved from Mathematic website: http://www.mathematica-mpr.com/~/media/publications/PDFs/education/playworks_brief1.pdf

Carlson, S.M., White, R.E., & Davis-Unger, A.C. (2014). Evidence for a relation between executive function and pretense representation in preschool children. *Cognitive Development, 29*, 1-16.

Hofferth, S.L. & Sandberg, J.F. (2001). Children at the millenium:Where have we come from, where are we going? *Advances in Life Course Research, 6*, 193-229.